COS 598/498: Generative AI Agents — Project Specification

Agent: PC Pal — Friendly IT Support & Digital Literacy Agent
Spring 2026 · University of Maine

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1. The Problem

What are you trying to do?

Step 1 — State the problem. Elderly and beginner computer users regularly feel lost, embarrassed, and helpless when they encounter routine technical problems or need to learn basic PC functionality. This creates a cycle of dependency on others (family members, IT help desks) rather than building lasting digital confidence and self-sufficiency.

Step 2 — Concrete example. Consider a 72-year-old retired teacher who needs to attend a telehealth appointment via Zoom, but doesn't know how to change her microphone input in System Settings. She calls her son, who talks her through it over the phone, but she still doesn't understand *why* she clicked what she clicked. The next week the same problem happens again. She can't see what her son is pointing at, he gets frustrated, and she feels ashamed. Nothing was actually *learned*.

Or consider a first-generation college student who has only ever used a smartphone. When a professor asks them to submit a PDF, they don't know how to take a screenshot, locate a downloaded file, or copy-paste error messages when asking for help. They feel too embarrassed to ask a classmate "basic" questions.

Step 3 — Prior solutions.

- **YouTube tutorials** (e.g., "How to change Zoom microphone settings") offer step-by-step video guidance, but they are static, not personalized, and require the user to already know how to find and navigate them.
- **Phone/chat IT help desks** (e.g., university IT support, Geek Squad) solve the immediate problem but do so *for* the user, not *with* them, building no durable skill.

Step 4 — Gap in prior solutions. Prior solutions fail to address the *root cause*: these users lack a patient, always-available guide who can meet them where they are, use their exact software/OS version, and teach skills in a way that builds genuine understanding — not just one-time fixes. YouTube is not interactive. Help desks don't teach. Neither approach adapts to the individual's pace, vocabulary, or current screen state.

Step 5 — Design question. How might a conversational AI agent, with access to annotated visual guides and the ability to ask clarifying questions, walk an elderly or beginner user through PC tasks step-by-step — building real digital literacy rather than just solving the immediate problem?

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2. Why Bother

Why is this problem important?

Over **57 million Americans are aged 65+**, and studies consistently show that the majority struggle with basic computer and internet tasks. Among college students, digital equity gaps disproportionately affect first-generation, low-income, and rural students who arrive without reliable prior tech exposure. As healthcare, government services, employment, and education all migrate online, the inability to navigate a PC is no longer a minor inconvenience — it is a barrier to medical access, financial stability, and academic success.

When this problem goes unresolved, users either disengage from technology entirely (with real health, civic, and economic consequences) or remain permanently dependent on others for tasks they could learn to do themselves — eroding their confidence and autonomy. The consequences are especially acute for elderly users navigating telehealth portals, and for beginners trying to succeed in college or find work.

AI agents are now well-positioned to help because they can generate personalized, step-by-step instructions on demand, ask clarifying questions ("Are you using Windows 10 or 11?"), and — critically — generate annotated visual guides that show users *exactly* what to click, with red circles and arrows highlighting UI elements. CollaborAITE's persistent channel history means the agent can remember that a user struggled with copy-paste two weeks ago and proactively reinforce that skill in new contexts.

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3. Status Quo

What are the current solutions and why do they fail?

Prior Solution Category 1: Static Video & Written Tutorials

Example: YouTube "How-To" Videos / Microsoft Support Articles

These resources offer detailed walkthroughs of common PC tasks. A user might search "how to take a screenshot Windows 11" and find a well-produced video or official Microsoft support page.

How users interact with it:

1. User encounters a problem
2. User must know to open a browser and search (already a barrier)
3. User watches/reads a generic tutorial not tailored to their OS version or situation
4. User must tab back and forth between the tutorial and the actual task
5. If the video doesn't match their screen (due to OS version differences), they're stuck

Where it falls short: Static tutorials cannot ask "What do you see on your screen right now?" They cannot adapt to whether the user is on Windows 10 vs. 11, or whether they already know what a "taskbar" is. There is no feedback loop. The user either succeeds silently or fails silently.

How our agent will differ: PC Pal asks clarifying questions before giving instructions, generates instructions calibrated to the user's exact environment, and can generate annotated screenshots with highlighted UI elements tailored to the step the user is currently on.

Prior Solution Category 2: Live Human IT Support

Example: University IT Help Desk / Geek Squad / Family Phone Support

Human support resolves the immediate problem through conversation, remote desktop access, or in-person assistance.

How users interact with it:

1. User encounters a problem and waits (often 20–45 minutes on hold)
2. Technician diagnoses and fixes the problem, often taking control of the machine
3. User watches but doesn't meaningfully participate
4. Problem is solved; no skill transfer occurs
5. Same problem recurs; cycle repeats

Where it falls short: Human support is expensive, time-limited, and optimized for *resolution*, not *education*. Technicians don't have time to explain why every step is taken. Users feel passive and dependent. Available only during business hours. Not scalable.

How our agent will differ: PC Pal is available 24/7, never impatient, always explains *why* each step matters ("We're going to the Settings menu because that's where Windows stores all the controls for your hardware — think of it like the control panel in a car"), and builds toward user independence rather than resolving the problem as quickly as possible.

Prior Solution Category 3: Existing AI Agents

Example: Microsoft Copilot (Windows integrated) / ChatGPT

General-purpose LLM assistants can answer IT questions conversationally and generate step-by-step instructions.

How users interact with it:

1. User opens ChatGPT or activates Copilot
2. User types a question (already a barrier for low-confidence users)
3. Agent generates a text-only response with numbered steps
4. User must interpret which on-screen element corresponds to the written description
5. No visuals, no memory of prior sessions, no proactive skill-building

Where it falls short: Current AI assistants output text instructions that assume visual familiarity with the UI. They do not generate annotated screenshots. They don't remember that this user needed help with copy-paste three sessions ago and hasn't fully internalized it yet. They don't teach; they answer. They can be verbose and technical, overwhelming novice users.

How our agent will differ: PC Pal combines conversational guidance with **generated annotated visual guides** (step-by-step images with red circles, arrows, and numbered callouts showing exactly which button to click). It uses CollaborAITE's persistent history to track which skills a user has learned and which need reinforcement, and it adjusts its vocabulary based on user profile (e.g., never saying "navigate to the registry" with a beginner).

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4. Your Proposed Agent

Field	Details
Agent Name	PC Pal
Description	Patient, visual IT tutor for elderly and beginner PC users
Team Name	[Your Team Name]
Team Members	[Names of 3–5 members]

4a. Agent Summary

PC Pal is a friendly, patient IT support and digital literacy agent designed specifically for elderly users and absolute beginners. Its core loop is: **the user describes a problem or task → the agent asks one or two targeted clarifying questions → the agent provides a step-by-step walkthrough with plain-language explanations AND auto-generated annotated visuals (screenshots or UI illustrations with red circles and numbered arrows) → the user follows along and asks follow-up questions in natural conversation → the agent reinforces the skill and checks for understanding.**

Addressing the "no visual guidance" gap: PC Pal generates step-by-step annotated image guides for every task — each image shows the relevant UI element highlighted with a red circle and a numbered callout that matches the instruction text. The user never has to guess "where is the Settings icon?" — it is circled in red in the image.

Addressing the "no skill retention" gap: Because PC Pal lives on CollaborAITE, it has access to the user's full interaction history. It tracks which skills have been introduced (e.g., copy-paste, screenshots, browser navigation) and weaves reinforcement into new interactions ("You'll use that same copy-paste skill here that we practiced last week — remember, Ctrl+C to copy!").

Addressing the "vocabulary mismatch" gap: PC Pal uses a vocabulary profile built from user interactions, avoiding jargon unless it has been explicitly introduced and explained. It always defines technical terms in plain English on first use ("The taskbar — that's the bar along the very bottom of your screen with the Start button").

4b. Data Sources

CollaborAITE Data Sources

- **Class slides and activities:** Not primary — but if deployed in a digital literacy course, slides on PC basics (file management, browsers, OS navigation) will be indexed so the agent can reference the curriculum the user is officially working through.
- **Channel conversations (including AI queries):** Core data source. The agent reviews prior @PC Pal conversations to identify which skills a user has asked about, which steps caused confusion (based on follow-up messages), and which topics have never been covered. This powers personalized skill-tracking.
- **User profile information:** Critical. At onboarding, the agent collects: OS and version (Windows 10/11, macOS), comfort level (never/rarely/sometimes/often uses a PC), specific goals (telehealth, email, schoolwork), and any accessibility needs (vision, motor). This shapes every response.
- **User-uploaded documents:** Users may share screenshots of error messages or confusing screens, which the agent uses to give pinpointed guidance ("I can see from your screenshot

that you're on Windows 10 — here's exactly where to find that setting on *your* screen").

- **Prior assignments and reflections:** If deployed in a digital literacy course, prior assignments reveal what skills have been formally taught, and reflections reveal what the user still finds confusing.
- **Anonymized session transcripts:** Used in aggregate to identify which tasks most commonly cause confusion for the user population, helping prioritize which visual guides to generate proactively.

Privacy-by-design limits:

- The agent does not access other users' private messages or assignments — only the current user's history — to protect individual privacy.
- The agent does not store or transmit screenshots of user screens to external services — all image analysis happens within the platform.
- The agent does not access OS-level data or installed application lists without explicit user share — respecting device privacy.

4c. Other Data Sources

Data Source 1: OS-Specific UI Screenshot Libraries

- *What it contains:* A curated library of annotated screenshots for Windows 10, Windows 11, and macOS Ventura/Sonoma covering the 50 most common beginner tasks (accessing Settings, taking a screenshot, copy-paste, browser navigation, file downloads, Zoom setup, etc.)
- *Why the agent needs it:* Generating accurate annotated visuals requires baseline screenshots of the correct OS version's UI. Without this, the agent would need to describe UI elements purely in text.
- *How it would be added:* Uploaded to the RAG knowledge base as a structured image library tagged by OS version and task category. The image generation sub-agent overlays red circles and arrows at specified coordinates.
- *Freshness:* Updated once per OS major release cycle (~annually). Maintained by the course instructor or a student assistant.
- *Licensing:* OS screenshots are generally permissible for educational fair use. Microsoft and Apple allow screenshots for instructional purposes.

Data Source 2: Plain-English IT Glossary

- *What it contains:* A structured glossary mapping technical IT terms to beginner-friendly definitions with analogies (e.g., "clipboard = a temporary invisible notepad that holds the last thing you copied").
- *Why the agent needs it:* Ensures consistent, accurate plain-language definitions every time a term is introduced. Prevents the agent from improvising definitions that could be confusing or inconsistent across sessions.
- *How it would be added:* Uploaded as a markdown document to the RAG knowledge base.
- *Freshness:* One-time upload; updated as needed when new OS features are introduced.
- *Licensing:* Created by the team; no external licensing issues.

4d. Human-Agent Collaboration Flow

```
[User] @PCPal I can't hear anything on my Zoom call
        |
        v
[PC Pal] Asks 1-2 clarifying questions
         "Are you using Windows or Mac? And do you see a speaker icon
          at the bottom of your screen?"
        |
        v
[User] Answers: "Windows, yes I see it"
        |
        v
[PC Pal] RAG lookup: OS = Windows 11, Task = Zoom audio
         Retrieves: annotated screenshot template + step sequence
         Sub-agent: generates annotated image with red circles
        |
        v
[PC Pal] Presents Step 1 in thread with annotated image
         "Step 1 of 4: Click the speaker icon (circled in red below) ↓
          [IMAGE: screenshot with red circle on speaker icon]
          Let me know when you've done this and I'll show you Step 2!"
        |
        v
[User] "Ok I clicked it, now I see a menu"
        |
        v
[PC Pal] Presents Step 2 with new annotated image
         Logs: user completed Step 1 successfully
```

```
        |
        v
... continues until resolved ...
        |
        v
[PC Pal] Skill wrap-up: "Great job! You just learned how to check
          your audio settings. Next time you can do this yourself.
          Want me to save a quick cheat sheet for you?"
        |
        v
[User] Optionally saves summary / marks task as learned
```

[Link to editable diagram: *add Figma/Miro/Lucidchart link here*]

4e. Example Interactions

Interaction 1 — "I can't copy and paste"

Scenario title: Teaching Copy-Paste from Scratch
Learner context: 68-year-old user, first week using a PC, trying to paste a password from an email into a login form.

```
@user: @PCPal I need to copy something from my email and put it in
       another window but I don't know how to do that

@PCPal: No problem at all — you're about to learn one of the most
        useful tricks on a computer! This is called "copy and paste."

        Quick question first: are you using a Windows computer or a Mac?
        (Windows usually has a Start button in the bottom-left corner.)

@user: Windows I think, yes I see the start button

@PCPal: Perfect! Here's Step 1 of 3.

        STEP 1 — Select the text you want to copy:
        Click at the very beginning of the text you want to copy.
        Then hold down your left mouse button and drag across the text
        until it turns blue (that means it's selected).

        [IMAGE: screenshot showing text highlighted in blue, with red
         circle around the highlighted area and arrow labeled "drag here"]

        Let me know when the text looks blue on your screen!

@user: ok its blue!

@PCPal: Excellent! You're doing great. Now Step 2:

        STEP 2 — Copy it:
        Hold down the "Ctrl" key (bottom-left of your keyboard, it says
        "Ctrl") and while holding it, press the letter "C" once.
        You won't see anything happen — that's normal! It's saved
        invisibly to your "clipboard" (think of it like your computer's
        temporary notepad).

        [IMAGE: keyboard diagram with red circle around Ctrl key and
         arrow to C key, labeled "press both at the same time"]

@user: done!

@PCPal: Perfect! Last step — Step 3:

        STEP 3 — Paste it:
        Click in the box where you want the text to appear (like the
```

```
        password box). Then hold "Ctrl" and press "V".
        Your text should appear!

        [IMAGE: screenshot of a login form with red circle around the
         password field, labeled "click here first, then Ctrl+V"]

@user: IT WORKED!! thank you so much

@PCPal: You did it! 🎉 Copy (Ctrl+C) and paste (Ctrl+V) are two of
        the most useful shortcuts on any computer. You can use them
        to copy text, pictures, file names — almost anything.

        I've saved a quick cheat sheet to your profile so you can
        look it up anytime. Want to practice once more or shall we
        move on?
```

Data sources used: User profile (Windows OS, beginner level), plain-English glossary (clipboard definition), keyboard diagram image library.

Interaction 2 — "How do I take a screenshot for tech support?"

Scenario title: Taking a Screenshot to Share an Error Message
Learner context: First-generation college student, getting a strange error when submitting an assignment. IT help desk told them to "send a screenshot."

```
@user: @PCPal the IT help desk told me to send them a screenshot of
       the error but I've never done that before, I don't even know
       what that means

@PCPal: A screenshot is just a photo of whatever is showing on your
        screen right now — super useful for tech support! Let me walk
        you through it. This will only take about 30 seconds.

        First: what kind of computer are you on?
        A) Windows (has a Start button at the bottom)
        B) Mac (has an Apple logo at the top left)

@user: Windows

@PCPal: Great! Here's the easiest way on Windows:

        STEP 1: Make sure the error message is visible on your screen
        (don't close it!).

        STEP 2: Press the Windows key + Shift + S at the same time.
        (The Windows key has the Windows logo — it looks like four
        small squares. It's usually between Ctrl and Alt at the bottom.)

        [IMAGE: keyboard with red circles on Windows key, Shift key,
         and S key. Arrow labeled "press all three at once"]

        Your screen will go slightly dark and a small toolbar will
        appear at the top. What do you see?

@user: I see a toolbar at the top with some icons

@PCPal: Perfect — that toolbar lets you select what to capture.

        STEP 3: Click the first icon (it looks like a rectangle with
        dotted lines — that's for selecting a region).

        [IMAGE: snip toolbar with red circle on rectangular snip icon]

        Then click and drag a box around the error message on your
        screen. When you let go, it's automatically copied!

@user: ok I drew a box around it
```

```
@PCPal: Great! It's now copied to your clipboard. To send it:

        STEP 4: Open your email to IT support, click in the message
        body, and press Ctrl+V to paste the screenshot right in.

        That's it! You just took your first screenshot. 📷

        Next time you see an error, you'll know exactly what to do.
        Want me to add "taking screenshots" to your skills list?
```

Data sources used: User profile (Windows, beginner), OS screenshot library (Windows Snipping Tool), prior interaction history (confirmed user knows Ctrl+V from Interaction 1).

Interaction 3 — System Settings: Changing Display Text Size

Scenario title: Making Text Bigger for Easier Reading
Learner context: 75-year-old user who mentioned in their profile they have some vision difficulty and is finding the screen text hard to read.

```
@user: @PCPal everything on my screen is so tiny, is there a way
       to make the writing bigger? my eyes aren't what they used to be

@PCPal: Absolutely, and this is one of the first things I recommend
        for anyone who finds the screen hard to read! We're going to
        go into your Display Settings and increase the text size.
        It's easier than it sounds, I promise.

        STEP 1: Click the Start button (the Windows logo in the
        very bottom-left corner of your screen).

        [IMAGE: desktop screenshot with large red circle around
         the Start button, arrow labeled "click here"]

@user: ok I clicked it, a menu popped up

@PCPal: Good! Now:

        STEP 2: Look for the word "Settings" — it has a gear icon ⚙️.
        Click on it.

        [IMAGE: Start menu with red circle around Settings gear icon]

@user: I see it, clicked it

@PCPal: You're doing great! The Settings window should be open now.

        STEP 3: Click on "System" — it should be near the top left.

        [IMAGE: Settings window with red circle around "System"]

@user: ok

@PCPal: STEP 4: On the left side, click "Display."

        [IMAGE: System settings left menu with red circle on "Display"]

        Then on the right side, look for "Scale" — it will show a
        percentage like "100%". Click the dropdown arrow next to it
        and try "125%" to start. Text and icons will get bigger!

        [IMAGE: Display settings with red circle on Scale dropdown,
         arrow pointing to 125% option]
```

```
@user: wow yes that's so much better!! why didn't anyone tell me
       about this sooner

@PCPal: So glad that helps! 😊 You can always go back to Display
        Settings to adjust it more if you want it even bigger —
        150% is also a popular choice.

        I've noted in your profile that you prefer larger text, so
        if you ever get a new device or reset settings, I'll remind
        you to do this first thing.
```

Data sources used: User profile (vision accessibility note, Windows OS), OS screenshot library (Windows 11 Settings), channel history (first interaction with this user).

4f. Harm Considerations

Potential Harm	Likelihood	Design Safeguard	What If It Fails?
Accuracy harm: Wrong instructions for user's OS version — agent gives Windows 10 steps to a Windows 11 user, causing confusion and distrust	Medium	Agent asks OS/version at onboarding and stores in profile; RAG library is tagged by OS version; agent confirms version before each task walkthrough	User gets confused and loses confidence; mitigation: agent asks "does this match what you see?" at each step and offers to try alternate instructions
Agency harm: Over-reliance / learned helplessness — user learns to always ask PC Pal rather than attempting tasks independently	Medium	Agent explicitly teaches *why* each step works, not just what to do; ends each interaction with a "skill unlocked" summary and encourages independent retry; tracks skill repetition and fades scaffolding over time	User doesn't develop independence; mitigation: agent proactively asks "want to try this one on your own first?" once a skill has been practiced 3+ times
Equity/accessibility harm: Instructions assume standard vision and motor ability — text instructions and annotated images don't work for users	Low-Med	Onboarding profile captures accessibility needs; agent can offer alternative input methods (voice-friendly	User cannot follow instructions; mitigation: agent always offers "would you like me to describe this in a different way?" and

Potential Harm	Likelihood	Design Safeguard	What If It Fails?
with significant visual or motor impairments		descriptions, larger-text instructions, keyboard-only alternatives to mouse actions)	flags the need for human IT support when the task is beyond accessible self-service

5. Technical Architecture

Platform: CollaborAITE (deployed as an @-mentionable agent in channels)
Language: Python
Agent Framework: Claude Agents SDK (for lower technical overhead) or Smolagents
Image generation: Sub-agent that takes (OS, task, step_number, UI_element_coordinates) → overlays red circles/arrows on base screenshot templates using PIL/Pillow
RAG database: CollaborAITE's built-in vector store, seeded with OS screenshot library + glossary + class materials

Major Modules:

1. **User Profile Manager** — reads/writes user profile document (OS version, skill history, accessibility needs, vocabulary level). Updated after every session.
2. **Task Classifier —** identifies the type of request (learn new skill / troubleshoot error / follow-up question / accessibility adjustment) to route to the right response strategy.
3. **Step Sequencer —** breaks tasks into numbered steps, presents each in a CollaborAITE thread, waits for user confirmation before advancing (paced walkthrough).
4. **Visual Guide Generator —** sub-agent that selects the correct base screenshot from the RAG library and renders red-circle/arrow annotations at specified coordinates using Pillow. Returns image as attachment.
5. **Vocabulary Filter —** checks proposed response against user's vocabulary profile; replaces flagged jargon with glossary-sourced plain-English alternatives.
6. **Skill Tracker —** logs which skills have been introduced/practiced; injects reinforcement reminders when a previously-learned skill is relevant to a new task.

Implementation Plan:

Week	Milestone
April 9–14	v1: Working agent with profile collection, text-only step-by-step instructions, basic task classification. No images yet.
April 14–16	v1 pilot with 2–3 users; collect feedback on tone, step clarity, vocabulary level
April 16–21	v2: Add Visual Guide Generator (annotated screenshots for top 10 tasks); add Skill Tracker; refine vocabulary filter based on feedback
April 21–23	v2 pilot with 2–3 users; observe whether visual guides reduce confusion
April 23–28	v3: Fade scaffolding logic; accessibility alternatives; full skill history dashboard for user
April 28	Final evaluation report; deployment recommendation

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6. Evaluation / Metrics of Success

6a. Success Criteria

Criterion 1: Users can successfully complete a new PC task on their own after one PC Pal walkthrough.
How you'd observe it: In the mini-evaluation, ask the user to attempt the same task again the following day without the agent. Observe whether they succeed independently or need to re-engage the agent.

Criterion 2: Users feel less anxious and more confident about using their PC after interacting with PC Pal.
How you'd observe it: 3-question pre/post survey ("On a scale of 1–5, how confident do you feel about [task]?"). Compare scores before and after session.

Criterion 3: The agent's instructions are accurate for the user's actual OS and UI (no hallucinated steps).
How you'd observe it: Team manually follows every agent-generated walkthrough on the corresponding OS version and verifies each step against the actual UI. Target: 0 errors in 10 randomly sampled responses.

6b. Design Workshop Plan

- **Demo:** Walk through Interaction 1 (copy-paste) live with a peer playing the "confused grandparent" role.
- **Show** a side-by-side of: text-only response vs. PC Pal's annotated visual response.
- **Questions for peers:**
  1. Is the tone genuinely patient and non-condescending, or does it still feel subtly patronizing?
  2. Are the step sizes right — or are single steps still too much for a true beginner?
  3. Does the visual guide actually make the task clearer, or is it confusing in its own way?
- **Capture:** Google Form with 5-point scales on the above + open comment field; one designated note-taker per group.

6c. Mini-Evaluation Plan

Participants: 2–3 users who self-identify as beginner/low-confidence PC users. Recruit from family members of team members, or via a digital literacy or introductory computing class on CollaborAITE.
Task: "Ask @PCPal to help you take a screenshot and email it to yourself. You've never done this before."
Observation method: Live observation (screen-share or in-person) + review of CollaborAITE conversation log afterward.
Feedback collection: 5-minute post-session interview:

1. Was anything confusing? Where did you get stuck?
2. Did the pictures help or were they distracting?
3. Do you think you could do this again on your own?
4. Did the agent ever feel impatient, rushed, or hard to understand?
   Analysis: Look for: steps where users stalled or asked follow-up questions (map to module fixes); vocabulary the agent used that users didn't understand; whether the annotated images were looked at or ignored.

6d. Final Evaluation Plan

Using collected interaction logs from all v1 and v2 pilot sessions, the team will:

- Calculate **task completion rate** (user reaches end of walkthrough without abandoning)
- Calculate **re-engagement rate** (same user returns to ask about the same task within 2 weeks — indicates skill was NOT retained)
- Analyze **follow-up question patterns** to identify which steps consistently cause confusion → feed back into step redesign
- Run a **vocabulary audit**: flag any responses where the user's reply suggests they didn't understand an agent term (e.g., "what does that mean?")
- Compare pre/post confidence survey scores across all participants
- Write a deployment recommendation addressing: Does PC Pal actually build independence, or does it create a new form of dependency?